



Bricks

A LEGO® BRICK MODEL OF

Nautilus Crew

Captain Nemo, Ned Land, Professor Aronnax, Conseil and a Nautilus diver on the deck of the Nautilus, in front of a salon window ringed with yellow lights.

106 pieces · 5 minifigures · 47 steps · 7 sections

Unofficial fan-made building instructions

Not physically build-tested. Every part is a real LEGO element and the model was checked on a computer (connections, collisions, every step, balance), but nobody has built it with real bricks yet.



Before you start

An unofficial fan-made manual

This booklet is not a LEGO product. It was generated from the model file by a computer program and checked part by part.

Not physically build-tested

The model and these steps were checked on a computer only: every part/colour is a real LEGO element, all 528 connections between parts were matched in 3D, no two parts overlap, every part can be pushed into place in its step, and the model's balance was estimated. Not tested: how firmly parts hold, part tolerances and how it copes with handling. Take your time and support the model while you work.

The finished model

About 12.8 × 6.4 cm and 9.3 cm tall, about 139 g (estimated).

Four minifigures of the main characters of Jules Verne's 20,000 Leagues Under the Sea, dressed as in Walt Disney's 1954 film, and one of Captain Nemo's divers in a brass deep-sea helmet, on a slice of the Nautilus's deck. Each figure is built from what LEGO sells: a printed head, a torso with its arms and hands, hips with legs, headwear and something to hold. The deck is riveted iron plating in reddish brown with a dark walkway, a round hatch and a mooring bollard, and a gold-trimmed edge with a blank black nameplate. Behind the crew a bulkhead carries one of the Nautilus's salon windows: a gold round frame, clear glass and a ring of eight yellow lights.

You will need

- The 106 pieces in 13 colours listed at the back, with their LEGO element IDs and BrickLink numbers.
- A brick separator for pieces that sit too deep for a finger.

Tips

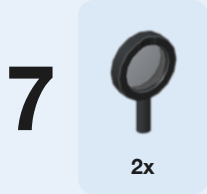
- Sort the pieces by colour before you start.
- Build the sections in order: each sub-assembly is finished before it is used.
- New pieces in a step have a yellow outline; earlier pieces are drawn paler.
- The figures' torsos come with their arms and hands already fitted, and their hips with the legs: you never take a figure apart to build it.
- Ned Land's harpoon, Nemo's spyglass, Aronnax's magnifying glass, Conseil's bottle and the diver's speargun clip into the figures' hands. Turn the arms and heads as you like: the pictures show one way to pose them.
- The diver's brass helmet (10165c01, from the 2012 Collectible Minifigures Series 8 Diver) has no model in the LDraw library: the pictures show brickkit's approximation of it. Its suit's print isn't modelled either, so the pictures show the torso and legs plain.
- The nameplate is a plain black 2 x 4 tile: add a printed tile or a sticker with the name if you like.

The 13 colours

 Reddish Brown: 58	 Medium Nougat: 5
 Yellow: 15	 Dark Brown: 3
 Black: 12	 Trans-Clear: 3
 Pearl Gold: 11	 White: 3
 Trans-Yellow: 8	 Dark Blue: 3
 Dark Bluish Gray: 7	 Flat Silver: 1
 Light Bluish Gray: 7	

How to read the steps

Parts and step number



The parts for the step and how many of each, then the step number. All parts are drawn at one scale.

New parts

The parts added in a step have a yellow outline and are drawn in full colour; the parts from earlier steps are drawn paler.

Sub-assemblies



A yellow-framed picture means: put in the sub-assembly you built in its own section. "Build 4x" means you need four of them.

Sections

Each section starts with its name at the top of the page. Sections are built in the order they appear.

Seen from below

When a step adds parts underneath, the picture shows the model from below. A small label says so.

Insert direction

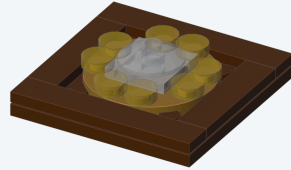
Some parts slide in sideways (axles through bearings, gears onto axles). The picture shows where they end up; push them in along their axle.

Build overview

The model is built in 7 sections. Each is finished before it is used.



1. Nautilus Crew
steps 1–12, 17, 23, 29, 35, 41, 47



2. Salon window
steps 13–16



3. Conseil
steps 18–22



4. Professor Aronnax
steps 24–28



5. Captain Nemo
steps 30–34

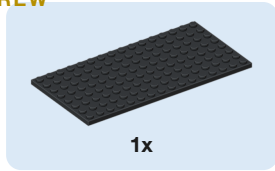


6. Nautilus diver
steps 36–40

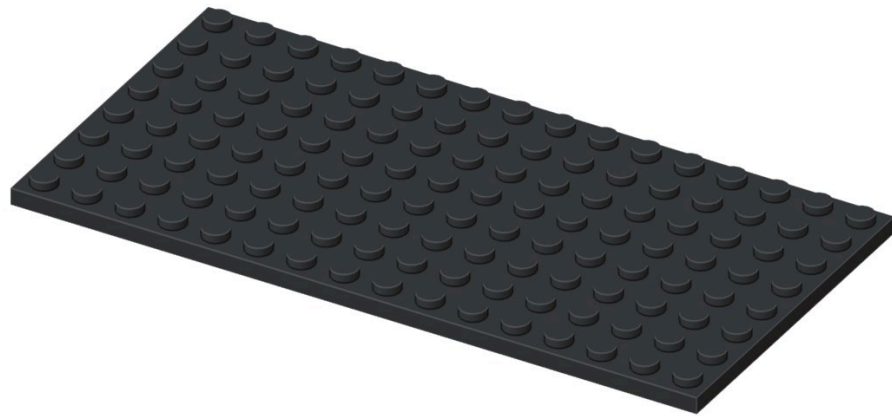


7. Ned Land
steps 42–46

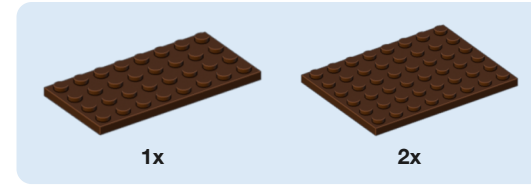
1



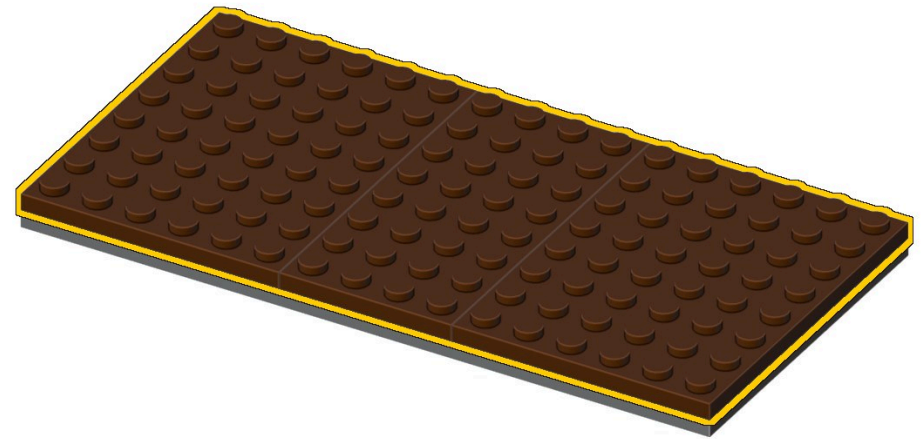
The bottom plate



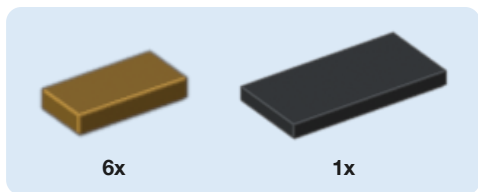
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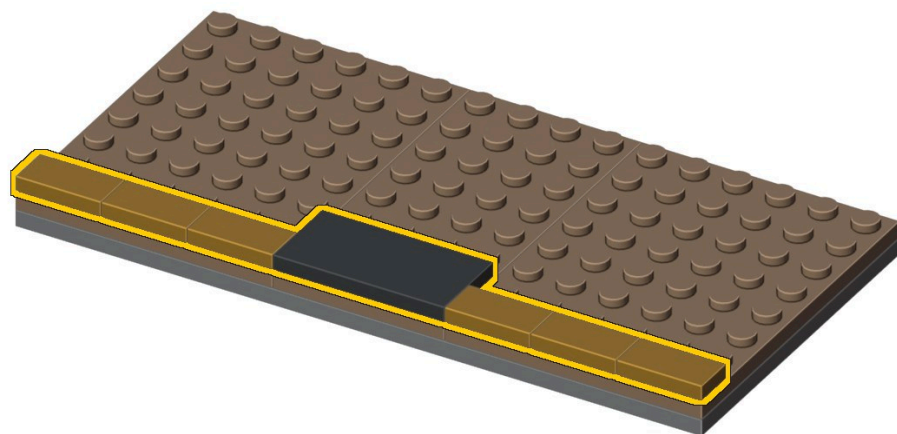
Three deck plates bridge it



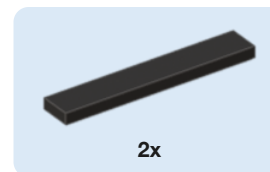
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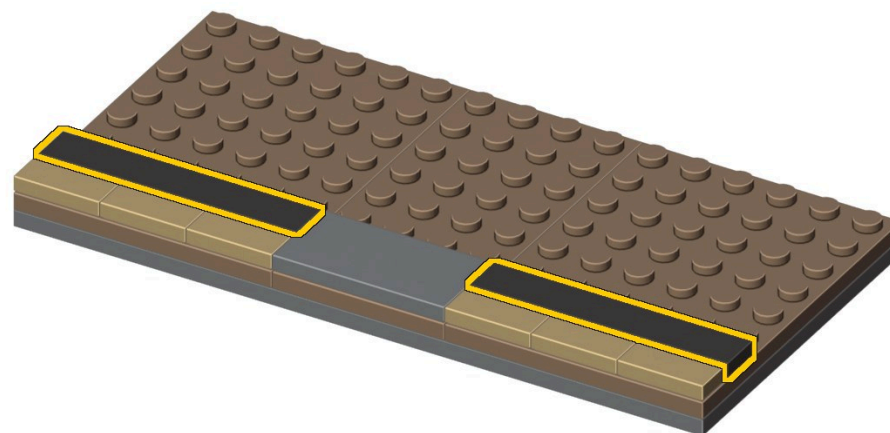
The gold edge and the nameplate



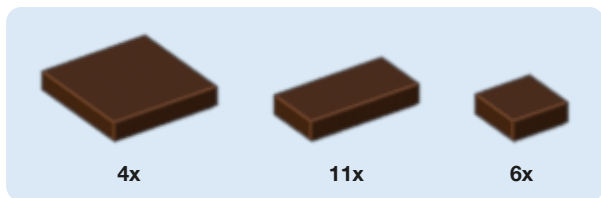
4



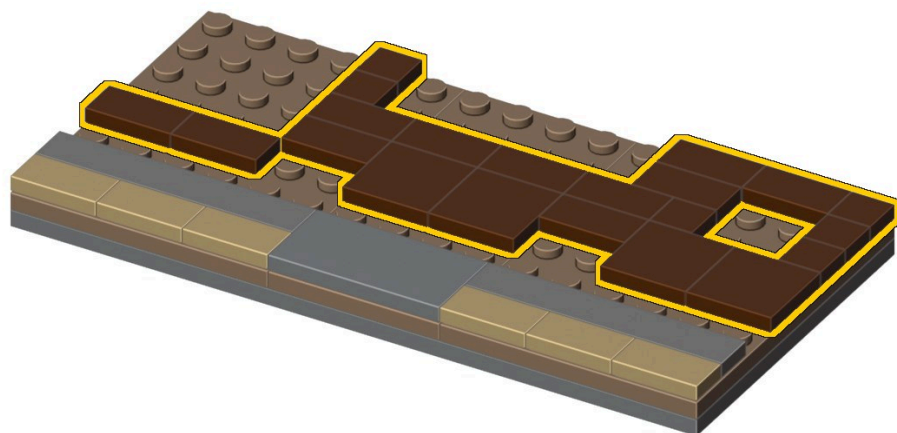
The walkway



5



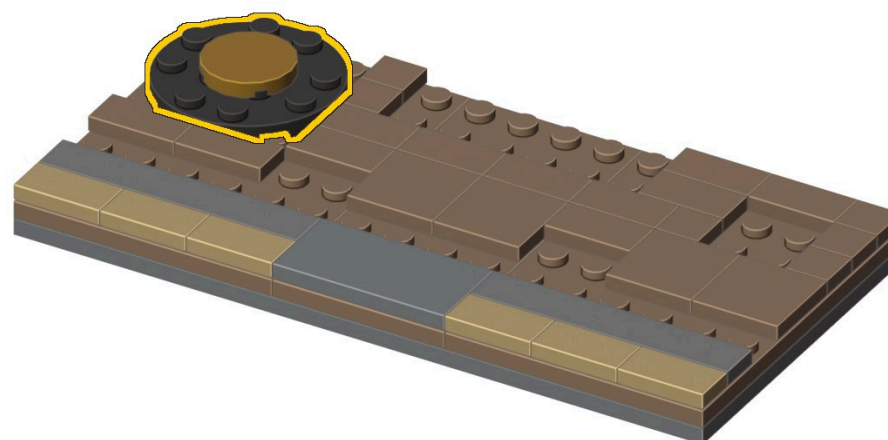
The deck plating



6



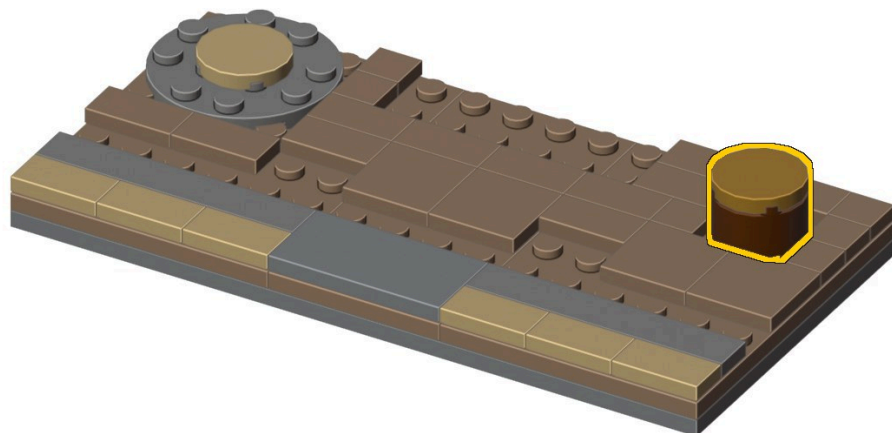
The hatch



7



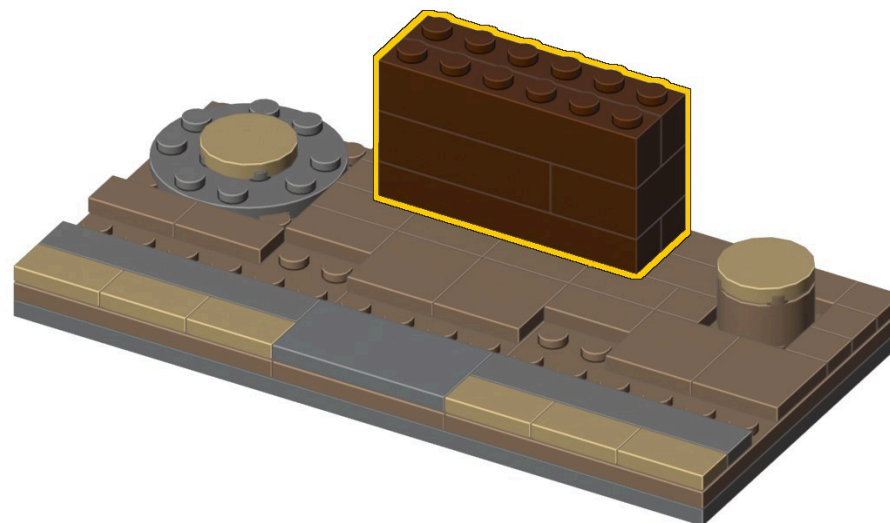
The mooring bollard



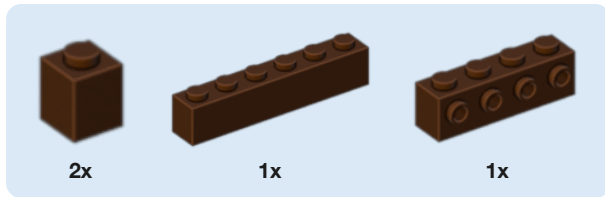
8



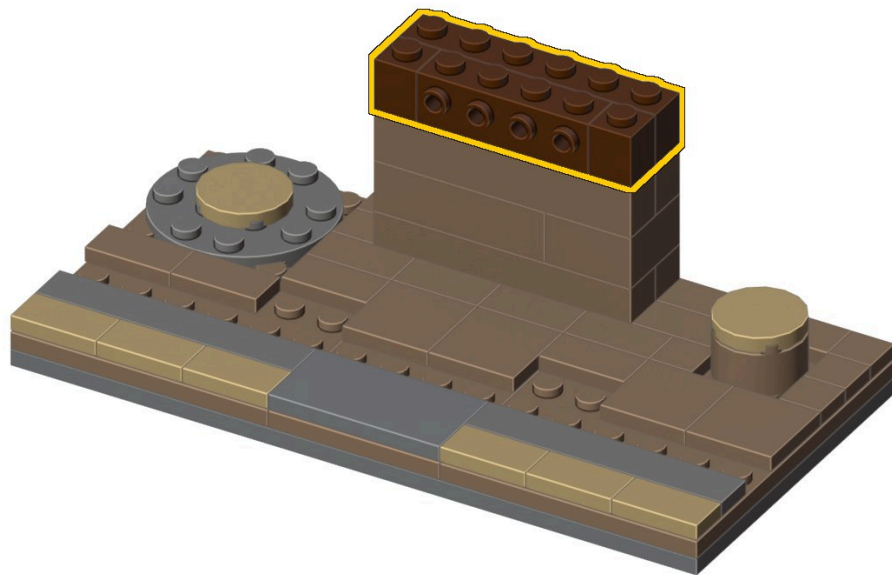
The bulkhead



9

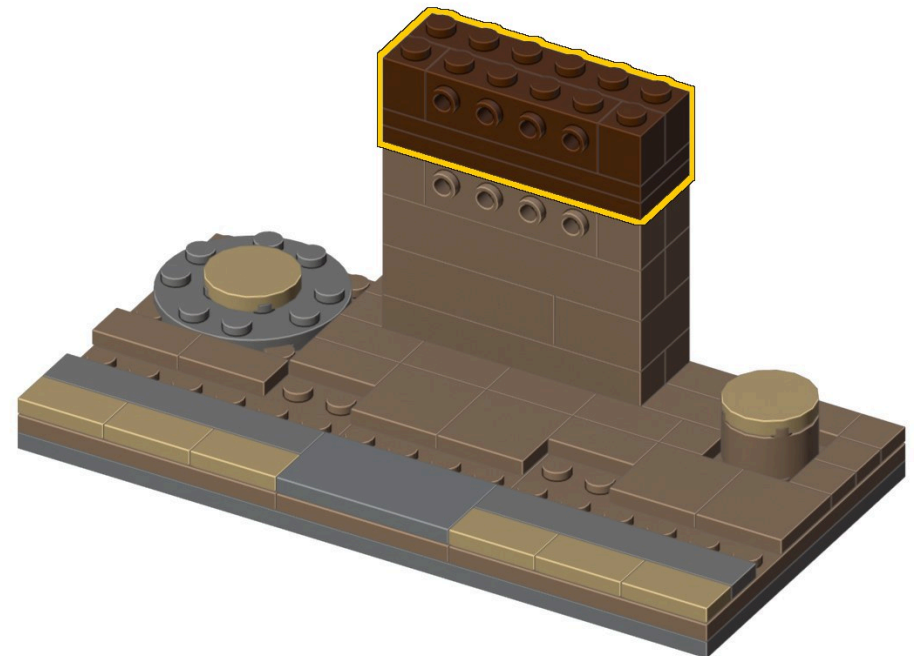
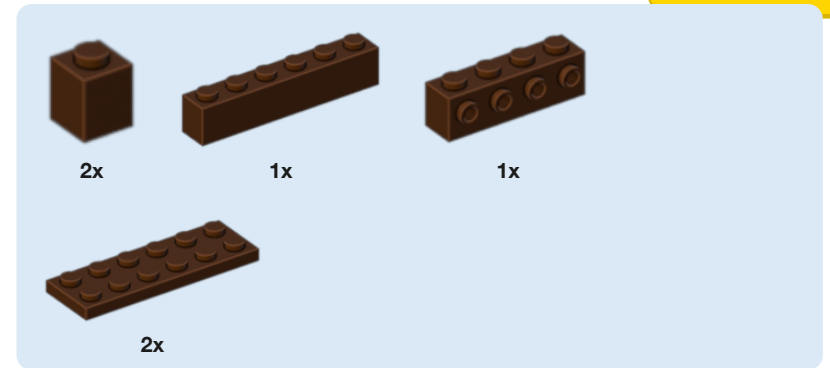


Studs on the side face the front

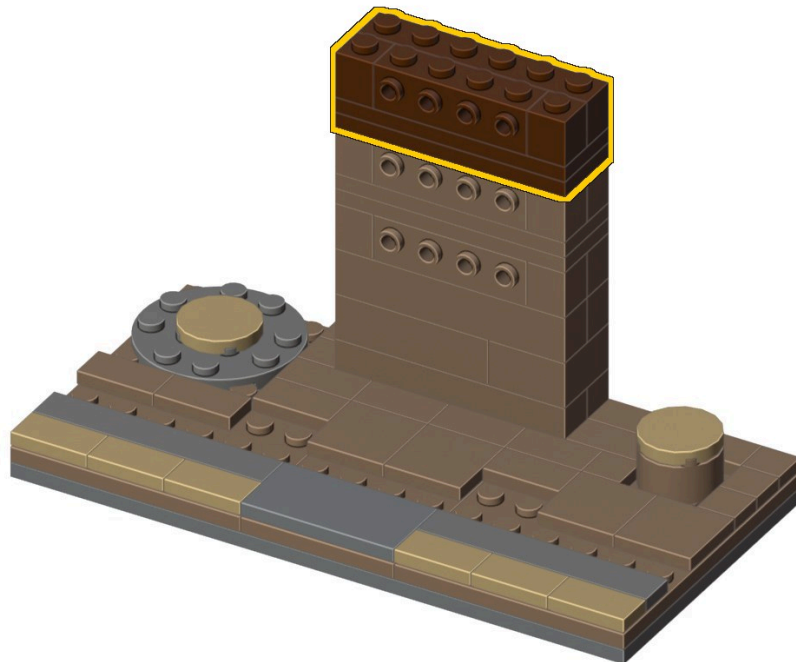
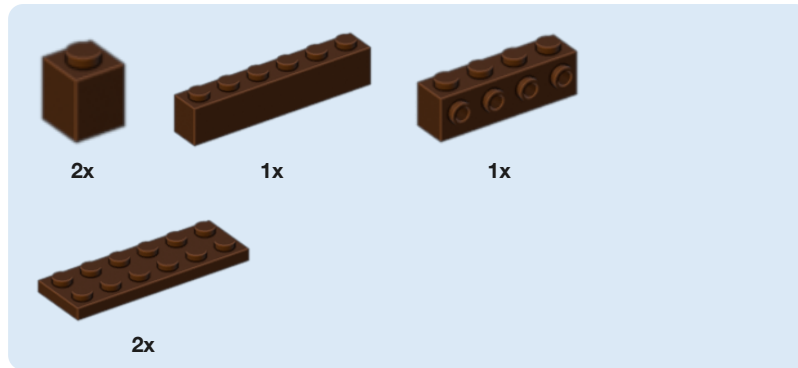


10

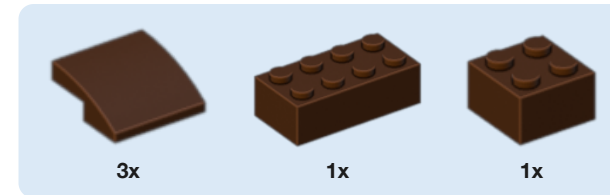
Nautilus Crew



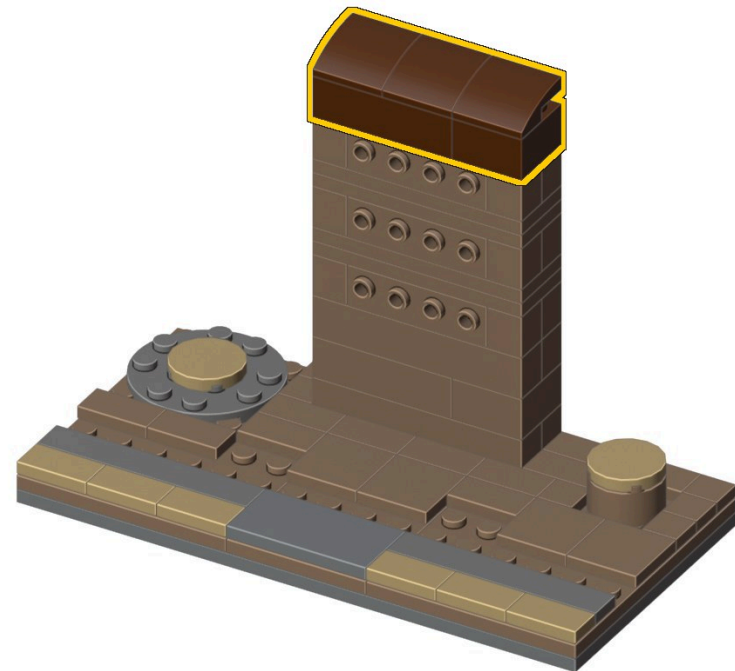
11



12



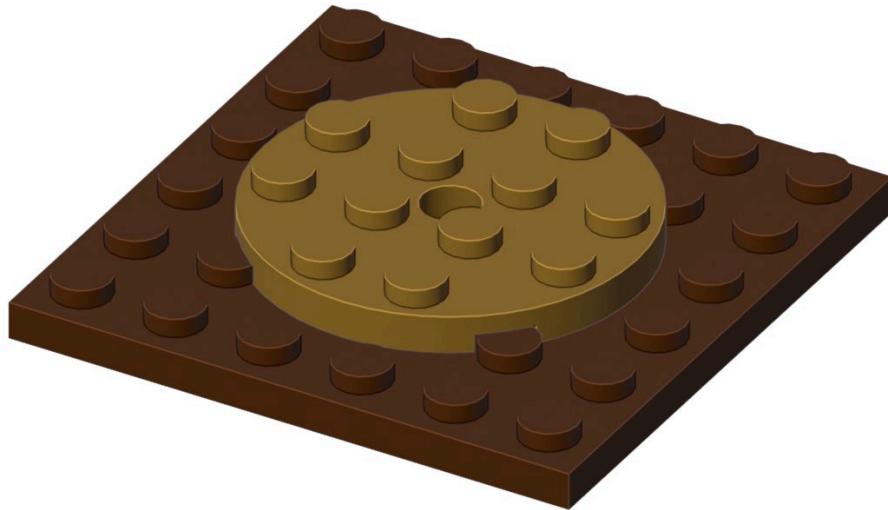
The bulkhead's rounded top



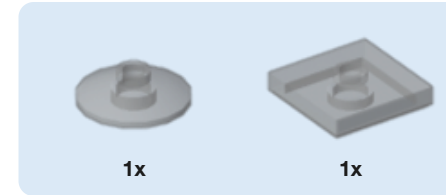
13



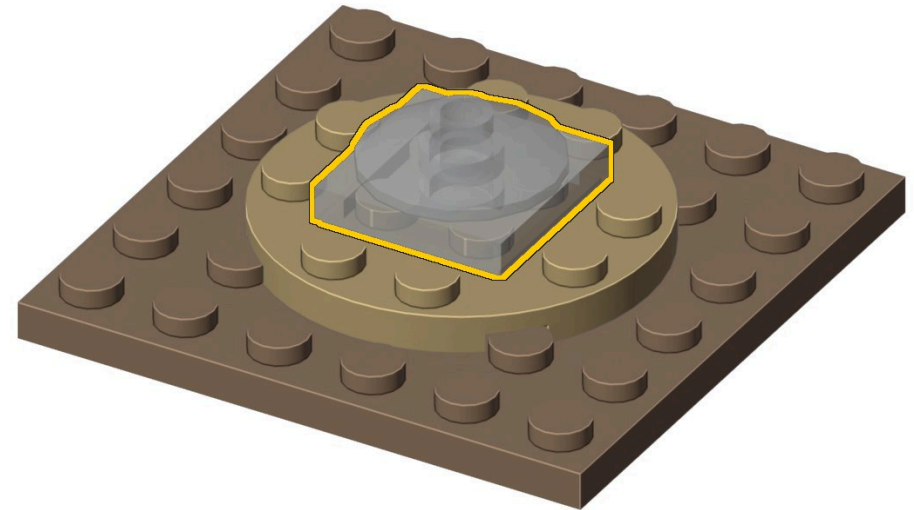
A gold round frame on the bulkhead plate



14



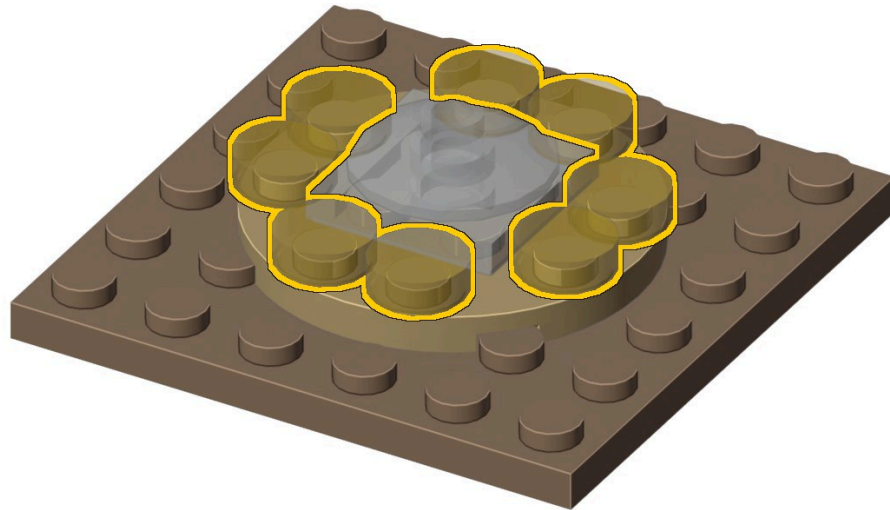
The glass



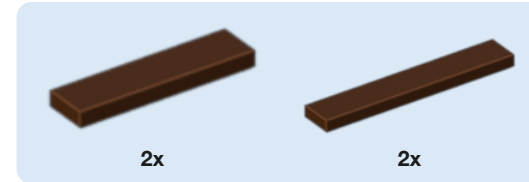
15



Eight yellow lights round the window



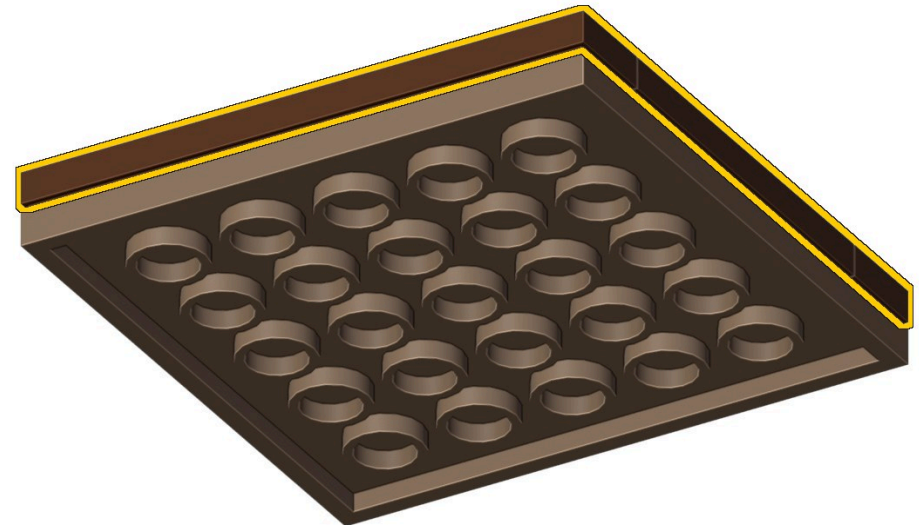
16



Plating round the frame (its corner studs stay: rivets)

Salon window

seen from below

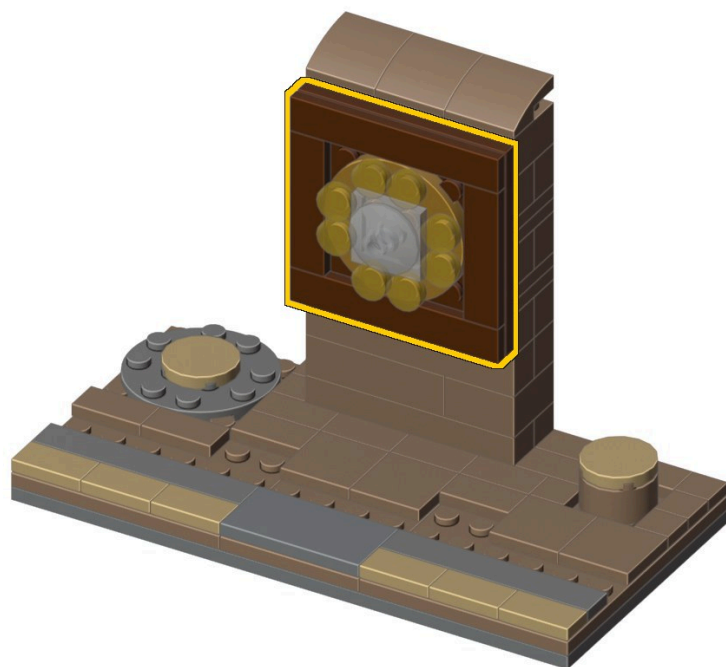


NAUTILUS CREW

17

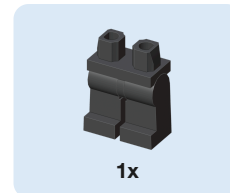


Press the window onto the side studs

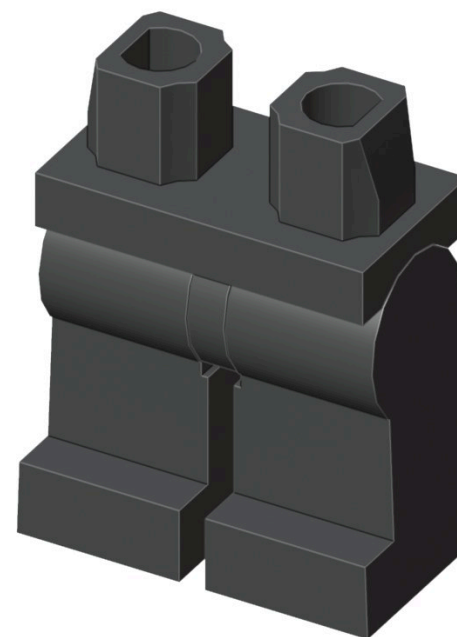


CONSEIL

18



Legs

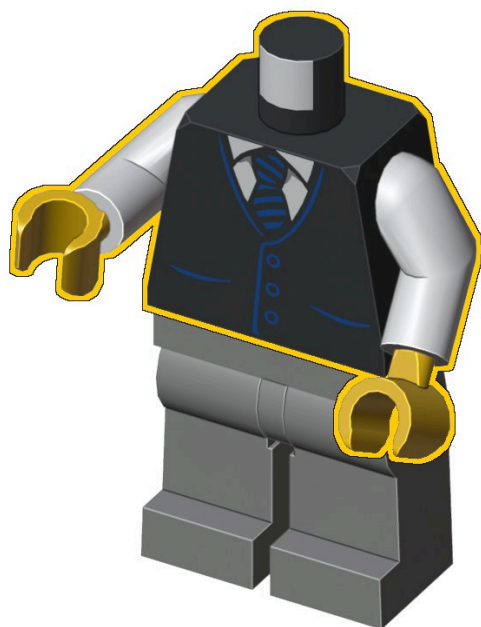


Conseil

19



Torso



20



Head



Conseil

21



Hat



22



Into the right hand



Conseil

NAUTILUS CREW

23

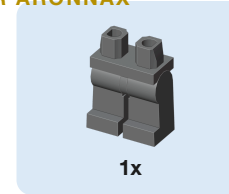


Conseil takes his place

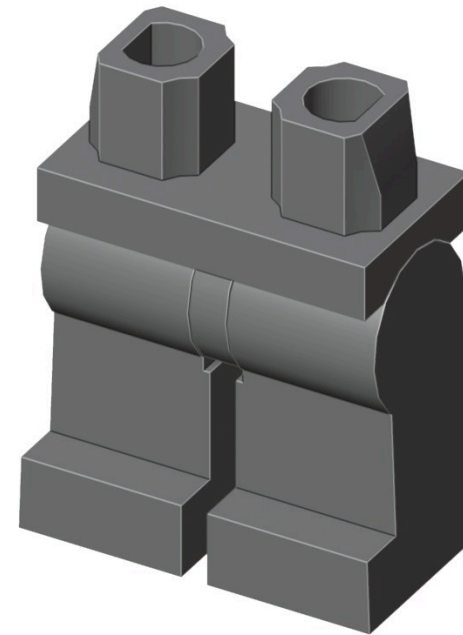


PROFESSOR ARONNAX

24



Legs

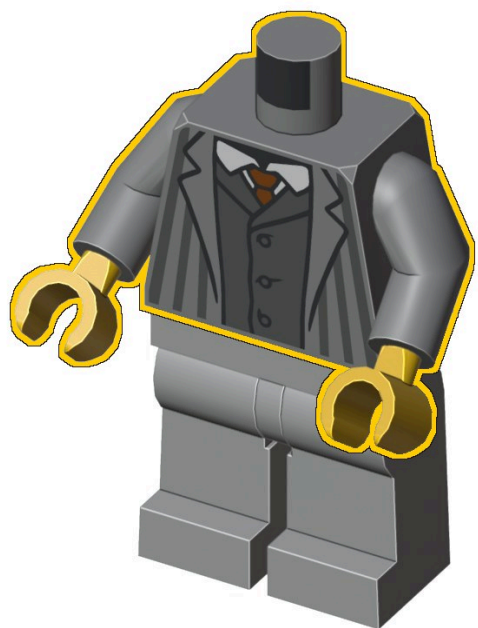


Professor Aronnax

25



Torso



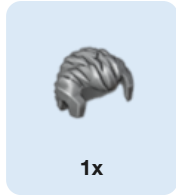
26



Head



27



Hair



28



Into the right hand



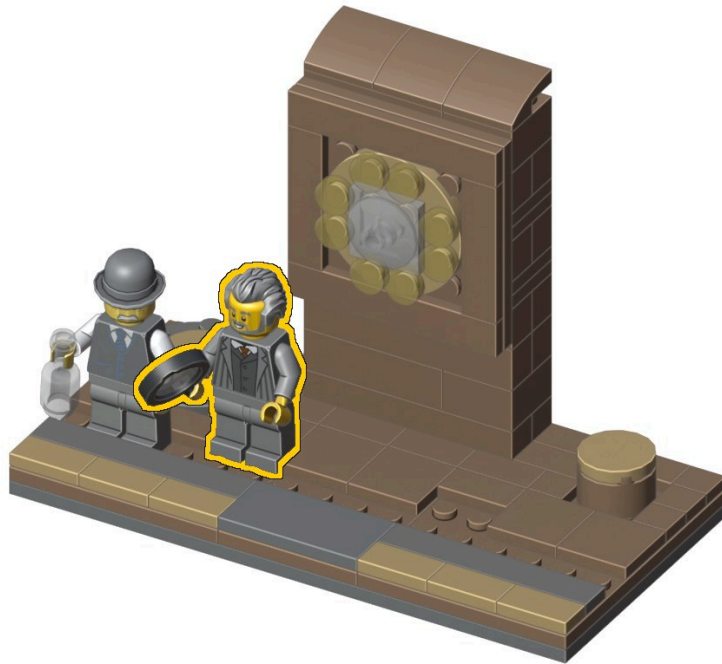
Professor Aronnax

NAUTILUS CREW

29

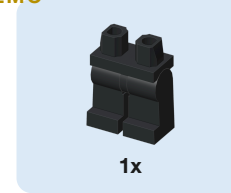


Professor Aronnax takes his place

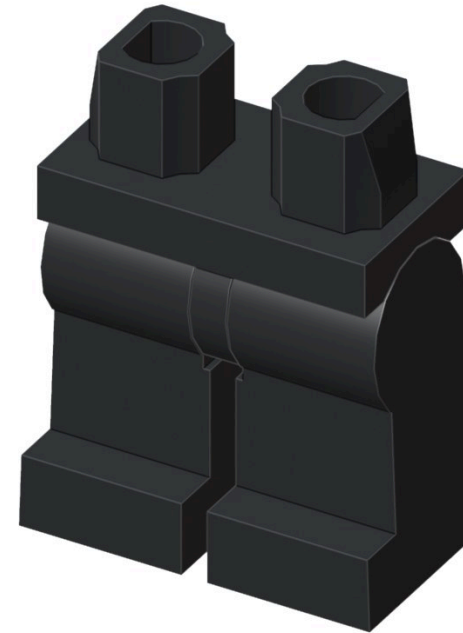


CAPTAIN NEMO

30



Legs



Captain Nemo

31



Torso



32



Head



33



Hair



34



Into the right hand



Captain Nemo

NAUTILUS CREW

35



Captain Nemo takes his place

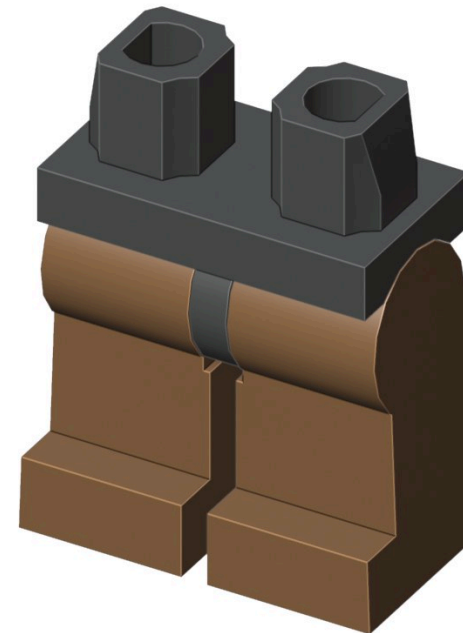


NAUTILUS DIVER

36

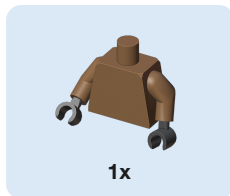


Legs

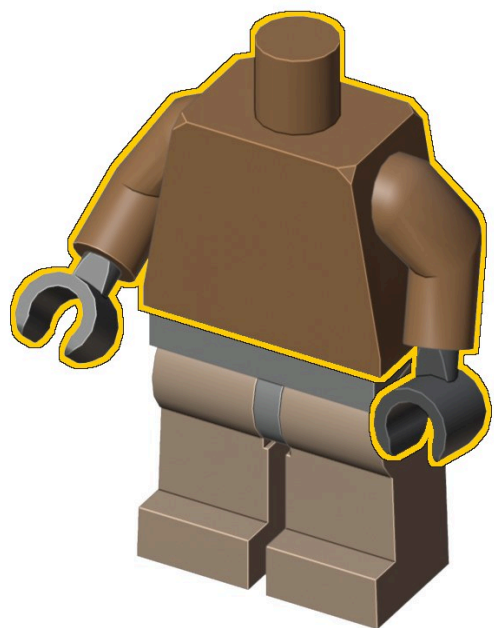


Nautilus diver

37



Torso



38



Head



39



Hat



40



Into the left hand



Nautilus diver

NAUTILUS CREW

41

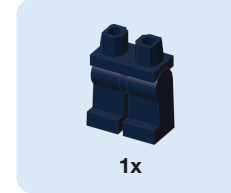


Nautilus diver takes his place

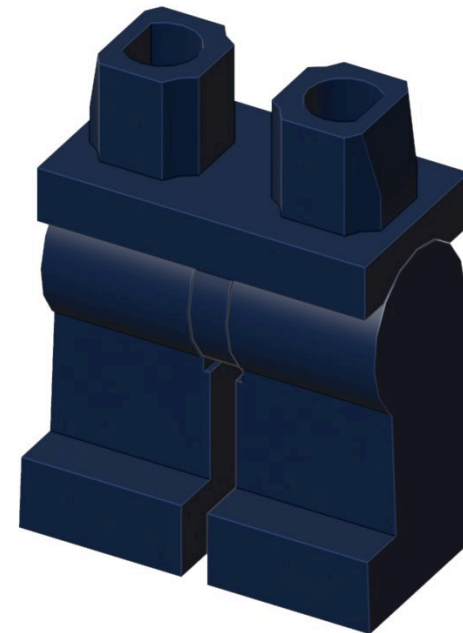


NED LAND

42



Legs



Ned Land

43



Torso



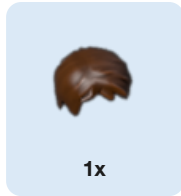
44



Head



45



Hair



46



Into the left hand





Ned Land takes his place



It works!

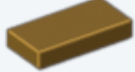
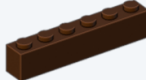
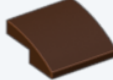
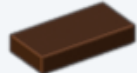


Real printed figures

Every head, torso and pair of legs is a real LEGO part with its print, listed by its BrickLink and Rebrickable number: no custom prints or stickers. Ned Land's striped shirt, Nemo's double-breasted captain's coat with its gold anchor and Aronnax's three-piece suit are all real LEGO torsos.

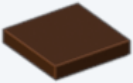
Parts inventory

106 pieces in 51 part/colour lines. Bold: the LEGO element ID (what LEGO Pick a Brick sells); below: the colour and the BrickLink part number.

 1x 6012466 Black BrickLink 10830c01	 1x 6414524 Black BrickLink 95674	 1x 4569093 Black BrickLink 970c00	 1x 6094066 Black BrickLink 92438	 1x 4560182 Black BrickLink 87079	 1x — Black BrickLink 973pb0049c01 <i>rare / discontinued</i>	 1x 6219624 Black BrickLink 973pb3159c01	 1x 4287672 Black BrickLink 30088
 1x 4569107 Dark Blue BrickLink 970c00	 1x 6359369 Dark Bluish Gray BrickLink 92081	 1x 6168713 Dark Bluish Gray BrickLink 970c00	 1x 6007953 Dark Bluish Gray BrickLink 970c150pb02 <i>rare / discontinued</i>	 1x 6005030 Dark Brown BrickLink 60474	 2x 6295366 Dark Brown BrickLink 6636	 1x 6162578 Flat Silver BrickLink 18041	 1x 6123037 Light Bluish Gray BrickLink 21268
 1x 6154364 Light Bluish Gray BrickLink 970c00	 1x 6468371 Light Bluish Gray BrickLink 973pb4899c02	 1x 6008374 Medium Nougat BrickLink 973pb1240c01 <i>rare / discontinued</i>	 1x 4568126 Pearl Gold BrickLink 64644	 1x 6018281 Pearl Gold BrickLink 10165c01 <i>rare / discontinued</i>	 1x 6521234 Pearl Gold BrickLink 60474	 6x 6107197 Pearl Gold BrickLink 3069b	 2x 6066345 Pearl Gold BrickLink 14769
 6x 4211242 Reddish Brown BrickLink 3005	 7x 4211193 Reddish Brown BrickLink 3009	 2x 4211210 Reddish Brown BrickLink 3003	 2x 4211201 Reddish Brown BrickLink 3001	 3x 6146858 Reddish Brown BrickLink 15068	 1x 4211166 Reddish Brown BrickLink 3941	 3x 6153594 Reddish Brown BrickLink 30414	 1x 6227107 Reddish Brown BrickLink 62810
 4x 4211247 Reddish Brown BrickLink 3795	 1x 4211207 Reddish Brown BrickLink 3035	 1x 4217848 Reddish Brown BrickLink 3958	 2x 4223729 Reddish Brown BrickLink 3036	 6x 4211288 Reddish Brown BrickLink 3070b	 11x 4211151 Reddish Brown BrickLink 3069b	 2x 4211194 Reddish Brown BrickLink 2431	 2x 4211204 Reddish Brown BrickLink 6636

Parts inventory (continued)

106 pieces in 51 part/colour lines. Bold: the LEGO element ID (what LEGO Pick a Brick sells); below: the colour and the BrickLink part number.

 4x 4271949 Reddish Brown BrickLink 3068b	 1x 6514246 Trans-Clear BrickLink 4740	 1x 6507876 Trans-Clear BrickLink 95228	 1x 6499361 Trans-Clear BrickLink 87580	 8x 6514193 Trans-Yellow BrickLink 98138	 1x 6316914 White BrickLink 973pb3974c01	 1x 6662107 Yellow BrickLink 3626pb1879	 1x 6218310 Yellow BrickLink 3626pb1965
 1x 4187305 Yellow BrickLink 3626px132 rare / discontinued	 1x 6218235 Yellow BrickLink 3626pb2199	 1x - Yellow BrickLink 3626pb0495					

About this model

Four minifigures of the main characters of Jules Verne's 20,000 Leagues Under the Sea, dressed as in Walt Disney's 1954 film, and one of Captain Nemo's divers in a brass deep-sea helmet, on a slice of the Nautilus's deck. Each figure is built from what LEGO sells: a printed head, a torso with its arms and hands, hips with legs, headwear and something to hold. The deck is riveted iron plating in reddish brown with a dark walkway, a round hatch and a mooring bollard, and a gold-trimmed edge with a blank black nameplate. Behind the crew a bulkhead carries one of the Nautilus's salon windows: a gold round frame, clear glass and a ring of eight yellow lights.

Checked on a computer

WARN	real elements	51 part/colour combinations: 0 not real, 5 rare; 15 minifig component(s), 2 drawn as stand-ins; 1 with an approximate 3D model
PASS	connections	528 connections (438 stud, 35 pin, 30 kit, 20 bar, 5 clip); 1 separate piece(s)
PASS	collisions	0 overlapping pair(s) among 136 parts
PASS	buildability	105 insertions tried across 17 submodel(s); 0 problem(s)
PASS	stability	about 139 g; centre of mass inside the base (77% margin); tips at 40.6 deg
PASS	mechanism	no moving parts defined
PASS	electrics	no electrics in this model
PASS	technique	0 note(s)

Not yet built with real bricks

These checks catch most mistakes a designer can make, but not all: clutch power, part tolerances, how stiff a long axle is and how the model survives handling can only be judged with real bricks. If you build it, we would love to hear how it went.

Files

3D model (LDraw, opens in BrickLink Studio, LeoCAD and LDCad), parts lists for BrickLink and LEGO Pick a Brick, and an interactive 3D viewer:

bricks.superfun.games/m/nautilus_crew

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